

Recognition of Sleep-Wake Stages by Deep Takagi-Sugeno-Kang Fuzzy Classifier with Random Rule Heritage

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Abstract—Due to the existence of noise and/or uncertainty in available sleep-wake stage data, recognition of sleep-wake stages with an efficient feature extraction has been facing the following challenges: (1) how to handle these uncertainties; (2) how to have interpretable recognition; (3) how to make recognition of sleep-wake stages to have good testing performance (i.e., generalization capability). To circumvent these three challenges, a deep Takagi-Sugeno-Kang (TSK) fuzzy classifier with random rule heritage called Drrh-TSK-FC is proposed for recognition of sleep-wake stages by integrating enhanced generalization capability of deep structure in fully excavating sleep information hidden in sleep stage data, together with the excellence of fuzzy classifiers in handling uncertainty and providing interpretability in this study. Drrh-TSK-FC assures both enhanced classification performance and strong interpretability by stacking the original input space plus the randomly projected outputs of the previous TSK fuzzy sub-classifier into current sub-classifier in a deep learning way. Besides, it has three distinctive characteristics: (1) the cognitive behavior is mimicked that people often leverage the past experiences (i.e., rules) to solve new yet similar problems. (2) in order to tackle with the changing importance of features caused by the randomness in the generation of the input space for each TSK fuzzy sub-classifier, some randomly generated short fuzzy rules are used. (3) a linear combination of all TSK fuzzy sub-classifiers is taken to facilitate with possibly different importance of the features in each sub-classifier. Moreover, Drrh-TSK-FC can be quickly trained by our previous work—least learning machine (LLM) or its variant in this study. The experimental results on recognition of sleep-wake stage tasks indicate that except for strong interpretability, the proposed Drrh-TSK-FC still keeps at least comparable classification performance to the adopted comparative methods.

Index Terms—Recognition of sleep-wake stages, fuzzy rule heritage, deep and sub-classifier learning, importance of features, classification performance.

I. INTRODUCTION

SLEEP can supplement human energy in time, enhance their own resistance and immunity, and effectively promote the normal growth and development of the human body. A healthy sleep is also very important to protect people's mental health and normal psychological activities [1]. Therefore, the diagnosis and treatment of sleep disorders has become one of the urgent problems to be solved [1].

Recognition of sleep-wake stages is an important process in the diagnosis of sleep-related diseases. Sleep stages are mainly derived from polysomnography (PSG) measurements during several hours of sleep [2]. PSG mainly records many physiological signals such as electroencephalogram (EEG), electromyogram (EMG), electrocardiogram (ECG) and electrooculogram (EOG). Among the recognition of sleep-wake stages, EEG is also one of the most widely studied physiological signals [2]. In 1968, Rechtschaffen and Kales developed a reading standard (R&K) [3]. In general, most of the existing methods about sleep-wake stage recognition firstly annotate time intervals in the PSG for different sleep states, for example, six sleep states: *wake-up (W)*, *stage 1*, *stage 2*, *stage 3*, *stage 4* and *rapid eye movement (REM)* [4], [5]. Secondly, they generate the off-line stage-wake data by using different readings of EEG signals according to a R&K standard. Thirdly, they finish the recognition task by carrying out various classification techniques on the generated data. To date, typical methods include: (1) Salih Gunes's method [6] which integrated a feature weighted (KMCFW) data preprocessing strategy together with *K-means*+KNN to finish the corresponding recognition task; (2) C.A.Holzmann's method which simply takes an artificial neural network to recognize sleep/wakefulness states in a minute-by-minute offline manner [7]; (3) L.G.Doroshenkov's method [8] which utilizes a hidden Markov model (HMM) to provide a seemingly satisfactory diagnostic and therapeutic program for sleepiness; (4) Yuan Ye's method [9] which uses a deep fusion attention network to learn multi-channel bio-signals, quantify the importance of each biomedical channel and implement enhanced feature representations based on these channels [9]. More advances about stage-awake recognition can be seen in [10], [11], [12].

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Like the literature [13] and experimental evidences [13] indicate, since the collected EEG signals and accordingly sleep-wake stage data are more likely associated with noise and uncertainties to a certain extent, an efficient feature representation (in both the preprocessing step and possibly successive classification steps) and robust classification (especially having strong uncertainty-handling and generalization capability) should be simultaneously emphasized to handling sleep-wake stage efficiently. At present, although some existing works enjoyed their efficient preprocessing steps, they sometimes ignore an efficient feature representation during the classification procedure, and give almost no any effort for both robust classification and interpretability. In order to achieve robust and interpretable classification of sleep-wake stage data, as a preliminary attempt, we have the following considerations in this study: (1) taking fuzzy systems as a basic tool for sleep-wake stage data classification to share their strong uncertainty-handling capability and interpretability [14], [15]. By *interpretability* in this study, we mean that the IF-parts of fuzzy rules are linguistically interpretable [14], [15]. (2) utilizing short fuzzy rules to reflect the importance change of the features extracted after the preprocessing step and hence have an efficient feature representation during the classification procedure. (3) taking a deep stacked structure of fuzzy systems to assure both satisfactory classification performance and enhanced generalization capability [16], [17]; (4) using short rule heritage from previous fuzzy systems in the deep structure to mimic the cognitive behavior that people often leverage the past experiences (i.e., rules) to solve new yet similar problems. Based on these four considerations, we develop a novel deep fuzzy classifier for sleep-wake stage data. The main contributions of this study can be summarized as follows.

- 1) In order to solve a new problem effectively and quickly, people generally consider how to inherit relevant knowledge summarized from the experiences about solving similar problems. Due to the similarity of the input space between adjacent sub-classifiers in the proposed deep structure, it becomes natural desire for us to mimic such a cognitive behaviour in such a deep structure. In this study, the concept of random rule heritage is proposed as the first attempt to achieve this goal.
- 2) The proposed deep structure shares the advantages of both enhanced classification performance and interpretability of the state-of-the-art deep fuzzy structures [16], [17] based on the stacked generalization principle [23]. Besides, it has its creativity in the generation of the input data space at each layer (i.e., each TSK fuzzy sub-classifier or each sub-classifier for simplicity) [16], [17]. That is to say, in order to propagate the outputs of the previous sub-classifier into the next sub-classifier and simultaneously keep the diversity as much as possible between the sub-classifiers, the input data space of each sub-classifier consists of the original data space plus the data space generated from the outputs of both the previous sub-classifier and the randomly inherited rules. While deep XAI (Explainable AI) including deep fuzzy XAI [48], [49] has been becoming a hot topic in the computational intelligence field, owing to the use of random rule heritage,

such an interpretable deep structure as a more general deep framework for fuzzy XAI obviously has its promising potential for practical applications in the future.

- 3) Each TSK fuzzy sub-classifier of Drrh-TSK-FC is equipped with some random short fuzzy rules so as to facilitate with changing importance of features in the proposed deep fuzzy classifier. The use of random short fuzzy rules actually simplifies the proposed deep structure and hence makes it become more interpretable.
- 4) In order to improve the classification efficiency of the proposed classifier, the output of each TSK fuzzy sub-classifier is learnt quickly by our previous work—least learning machine (LLM) [18], [19] or its variant in this study. The experimental results on sleep-wake staging data show that except for strong interpretability, the proposed classifier has at least comparable classification performance to the baseline methods.

The remaining parts of this study are organized as follows. Section II briefly reviews related works for sleep-wake stage recognition. A novel deep changing-feature-importance-based TSK fuzzy classifier called Drrh-TSK-FC for recognition of sleep-wake stages is described in Section III. Empirical results of Drrh-TSK-FC are presented in Section IV. In Section V, we conclude this study.

II. RELATED WORKS

Now we summarize the existing works about sleep-wake stage recognition. Owing to today's prevalence of recently developed deep learning, here we call all the existing sleep-wake stage recognition methods rather than deep learning methods the traditional methods for easy understanding.

1) Traditional Recognition Methods for Sleep-Wake Stage

Since the early 1990s, although traditional automatic recognition of sleep-wake stages is a draining, inefficient and subjective evaluation process, it has become one of the hottest research topics. Such existing researches usually include the following steps: (1) pre-processing such as noise removal from original EEG signals; (2) extracting features; (3) developing an appropriate classifier to evaluate the acquired sleep-wake stage dataset. Traditional recognition methods roughly deal with the following technologies [1], [13], i.e., expert evaluation model [20], extreme learning machine (ELM) [37], hidden Markov models [21], random forest [10], [22], Linear Discriminant Analysis (LAD) [24], subject-specific classification model [25], features segmentation and self-organization (clustering) [38] and fast discriminative complex-valued convolutional neural network (FDCCNN) [1]. However, these methods sometimes do not yield desirable results for us. Besides, these existing methods do not break the following barriers: (1) Noise existing in current sleep monitoring systems makes it extremely difficult to extract their features accurately and even makes the importance of extracted features become changing over time [39]. As a result, the extracted features of sleep-wake stage data often fail to clearly represent features [40]. (2) There exist uncertainties in the sense of both the importance of each feature over time

and the choice of an appropriate feature representation method. As we know well, fuzzy system modeling has been proved to be an effective means for handling noises and uncertainties [28], therefore, fuzzy systems can naturally be a desirable choice in this study.

2) Deep Learning Methods for Sleep-Wake Stage Recognition

In contrast to the above traditional methods, deep learning methods for sleep-wake stage recognition are taken as the latest efforts in breaking through one or some of the above barriers. For example, Tuncer Seda Arslan [20] et al. developed a decision support system to determine sleep disorders. They identified sleep patients and healthy individuals only by pulse transit time (PPT). Their proposed method yields good classification results, especially using the deep learning method to successfully cope with the difficulty in an efficient feature extraction [20]. Junming Zhang [13] et al. proposed an automatic sleep stage method by constructing several sparse deep belief nets to extract features from sleep stage signal and hence improve the classification accuracy of sleep staging. Based on neurophysiological signals, Yudong Sun [41] et al. designed a deep convolution neural network model for automatic recognition of sleep-wake stages by successively increasing the depth, extracting the multi-level features and training the corresponding modules [41]. The developed model also yielded good classification results in the sense of the macro-averaging F1-score [41]. However, since these models behaves in black-box way, and they do not consider both the changing importance of the features and the uncertainty existing in sleep-wake stage data, their improvements are still being expected. Previous works [16], [17] reveals that deep stacked fuzzy classifiers can achieve good trade-off between promising performance and interpretability for a variety of classification tasks. Therefore, different from these existing deep learning methods, we will develop a novel deep stacked fuzzy classifier to classify sleep-wake stage data in this study.

III. A NOVEL DEEP TAKAGI-SUGENO-KANG FUZZY CLASSIFIER

In this section, we state the proposed deep fuzzy classifier Drrh-TSK-FC, including its training structure, its learning algorithm and its analysis. Because Drrh-TSK-FC consists of several TSK fuzzy classifiers (as the TSK fuzzy sub-classifiers) in stacked way, we first review extensively-used TSK fuzzy classifier in brevity.

A. TSK Fuzzy Classifier

The classical TSK fuzzy classifier is involved in this study. Now, their brief reviews are given in this subsection. A classical TSK fuzzy classifier is represented by the following fuzzy rules [26], [27].

The l -th fuzzy rule R^l :

IF x_1 is A_1^l x_2 is $A_2^l \wedge \dots \wedge x_d$ is A_d^l

Then

$$f^l(\mathbf{x}) = [p_0^l, p_1^l, \dots, p_d^l] \cdot [1, x_1, \dots, x_d]^T \quad l = 1, 2, \dots, L. \quad (1)$$

where A_j^l is a fuzzy set in the j -th input feature x_j for the l -th rule, $\wedge$ is a fuzzy conjunction operator and L is the number of fuzzy rules. Each rule is premised on the input vector $\mathbf{x} = [x_1, x_2, \dots, x_d]^T$. $f^l(\mathbf{x})$ is a linear function of the input vector $\mathbf{x}$ [26], [27]. The output of the TSK fuzzy classifier can be written as follows.

$$y^0 = \sum_{l=1}^L \frac{\prod_{j=1}^d u_{A_j^l}(x_j)}{\sum_{l'=1}^L \prod_{j=1}^d u_{A_j^{l'}}(x_j)} \sum_{j=0}^d p_j^l x_j \quad (2)$$

where $u^l(\mathbf{x})$ is the fuzzy membership function [26], [27]. We usually use the following Gaussian function as the fuzzy membership function [26], [27]:

$$u_{A_j^l}(x_j) = \exp\left(- (x_j - a_j^l)^2 / (2\sigma_j^l)^2\right) \quad (3)$$

In (3), we can get a_j^l and σ_j^l by some clustering techniques or other approaches [26], [27]. If fuzzy c -means (FCM) [30], [43] is adopted, a_j^l and σ_j^l can be expressed as [26], [27]

$$a_j^l = \sum_{i=1}^N u_{il} x_{ij} / \sum_{i=1}^N u_{il} \quad (4)$$

$$\sigma_j^l = h \cdot \sum_{i=1}^N u_{il} (x_{ij} - a_j^l)^2 / \sum_{i=1}^N u_{il} \quad (5)$$

where u_{il} denotes the fuzzy membership of the i -th input sample x_i and h is the scale constant given by users. For the classical TSK fuzzy classifier, researchers usually use two forms to represent the output $f^l(\mathbf{x})$. If $f^l(\mathbf{x}) = p_0^l$, we can call this TSK fuzzy classifier as zero-order TSK fuzzy classifier [16], [17]. If $f^l(\mathbf{x}) = \sum_{j=0}^d p_j^l x_j$, this kind of TSK fuzzy classifier can be called first-order TSK fuzzy classifier [16], [17]. We can easily write the output of the classical TSK fuzzy classifier [16], [17], [26], [27] as follows

$$y^0 = \sum_{l=1}^L \left(\prod_{j=1}^d u_{A_j^l}(x_j) \right) \sum_{j=0}^d p_j^l x_j \quad (6)$$

B. On the Deep Structure of a TSK Fuzzy Classifier

Drrh-TSK-FC is developed to design an ingenious deep learning structure for enhancing classification performance and interpretability for recognition of sleep-wake stages. Now we describe the structure of Drrh-TSK-FC and its learning algorithm and prediction procedure.

Here we state the design methodology of the proposed deep fuzzy classifier Drrh-TSK-FC in detail. Owing to practical demand of both interpretability and promising classification performance, we take the stacked structure [23] as the principal part of our design. This is because each TSK fuzzy sub-classifier has its high interpretability, and such a stacked structure indeed assures both interpretability and enhanced classification performance, according to stacked generalization principle [23]. In the proposed structure of Drrh-TSK-FC, the input space of each TSK fuzzy sub-classifier keeps the same as in [23]. That is to say, the original input space plus the randomly projected outputs of the previous TSK fuzzy sub-classifier is presented as

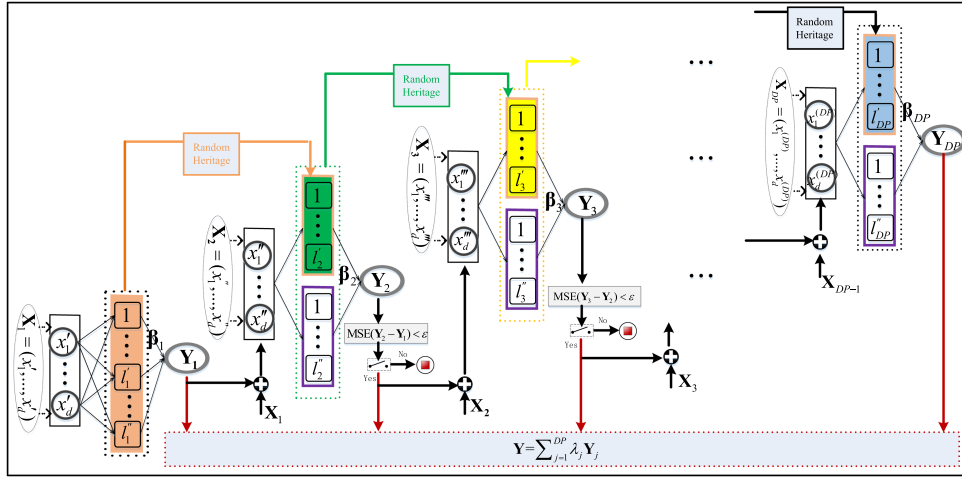


Fig. 1. The structure of Drrh-TSK-FC.

the input space to current TSK fuzzy sub-classifier [16], thereby resulting in possibly changing importance of the features caused by the use of random projection. In addition, due to the existence of noise, uncertainty and possible importance change of the extracted features over time for stage-wake recognition tasks, Drrh-TSK-FC facilitates with three particular contributions.

- 1) *the use of fuzzy rule heritage*: Fuzzy rule heritage reflects human's strong cognitive behavior in the sense of leveraging the past experiences to solve new yet similar problems. The proposed Drrh-TSK-FC propagates some randomly selected fuzzy rules from the previous sub-classifier into the next sub-classifier as the fuzzy rule heritage, i.e., a part of the set of all fuzzy rules in the next sub-classifier.
- 2) *the use of short fuzzy rules*: By *short fuzzy rules*, we mean the corresponding fuzzy rules adopt most of the features in their IF parts. Obviously, by ignoring some features, to some extent the use of *short* fuzzy rules can reflect the importance change of the extracted features over time. Some *short* fuzzy rules are randomly generated in each sub-classifier.
- 3) *the use of linear combination of sub-classifiers*: Due to the existence of possible importance change of the extracted features along each sub-classifier, Drrh-TSK-FC does not simply take the outputs of the final sub-classifier as its outputs. Instead, in order to achieve the guarantee of enhanced classification performance, Drrh-TSK-FC takes a linear combination of the outputs of all sub-classifiers as its outputs according to the same strategy as in [50].

Given the whole training dataset $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^T$ and the corresponding class label set $\mathbf{T} = [t_1, t_2, \dots, t_N]^T$. Let DP be the number of sub-classifiers, and K_{dp} as the number of fuzzy rules involved in the dp -th sub-classifier. Fig. 1 illustrates the structure of Drrh-TSK-FC. Now let us state how Drrh-TSK-FC is generated according to its stacked structure [23].

Obviously, the structure in Fig. 1 consists of both DP sub-classifiers and their linear combination. By simply running the classical clustering algorithm FCM [30], [43] to determine the IF-parts of fuzzy rules, the first sub-classifier can be generated.

In particular, we may use some randomly generated short fuzzy rules to facilitate with the importance change of the features in sleep-wake stage data. We also adopt FCM [30], [43] to determine two important parameters of Gaussian membership functions in the IF-parts of the corresponding fuzzy rules in the first sub-classifier. This can be readily done by using (4) and (5). Hence, below we mainly state the structures of the remaining sub-classifiers (see *Part I*), their linear combination (see *Part II*) and the deep learning algorithm (see *Part III*).

Part I—On the structure of the remaining sub-classifiers: In contrast to the generation of the first sub-classifier, the key difference is the use of random rule heritage in all the remaining sub-classifiers. Considering the very similarity between the generated input spaces of two adjacent sub-classifiers, we propagate some fuzzy rules from the previous sub-classifier into current sub-classifier so as to mimic human's cognitive behavior in solving similar problems. This induces the so-called *random rule heritage*. More concretely, for the dp -th sub-classifier ($dp \geq 2$), we first fix the number l_{dp} of fuzzy rules in which the total l'_{dp} fuzzy rules are randomly selected from the $(dp - 1)$ -th sub-classifier as *random rule heritage* and the remaining l''_{dp} fuzzy rules are generated by random generation strategy [16]. Such a random rule heritage aims at mimicking human's strong cognitive behavior in leveraging the past experiences to address similar problems, while the total l'_{dp} randomly generated fuzzy rules maintain both the classification performance and generalization for current input space. That is to say, the total l'_{dp} randomly inherited rules from the $(dp - 1)$ -th sub-classifier is first obtained as follows:

R^1 : IF x_1 is $\bar{A}_1^1 \wedge x_2$ is $\bar{A}_2^1 \wedge \dots \wedge x_d$ is $\bar{A}_d^1$,

Then $f^1(\mathbf{x}) = \sum_{j=0}^d p_j^1$

$\dots$

$R^{l'_{dp}}$: IF x_1 is $\bar{A}_1^{l'_{dp}} \wedge x_2$ is $\bar{A}_2^{l'_{dp}} \wedge \dots \wedge x_d$ is $\bar{A}_d^{l'_{dp}}$,

Then

$$f^{l'_{dp}}(\mathbf{x}) = \sum_{j=0}^d p_j^{l'_{dp}} \quad (7)$$

TABLE I
GENERATED DATASETS

Datasets	Polysomnia Fragments Combinations	Sizes of Samples	Features
Data-1	slp01a-slp01b-slp02a-slp02b-slp03	240-240-266-266-26	10
Data-2	slp04-slp14-slp16-slp32-slp37	293-266-333-266-26	10
Data-3	Slp41-slp45-slp48-slp59	266-266-266-266	10
Data-4	Slp60-slp61-slp66	266-266-266	10

where $\bar{A}_j^{l_1} = \bar{u}_{\bar{A}_j^{l_1}}(x_j) = \exp(-(x_j - \bar{a}_j^{l_1})^2 / (2\bar{\sigma}_j^{l_1})^2)$, $\bar{A}_j^{l_1}$ is a fuzzy set corresponding to the input feature x_j for the l_1 -th fuzzy rule, in which $\bar{a}_j^{l_1} = \sum_{i=1}^N u_{il_1} x_{ij} / \sum_{i=1}^N u_{il_1}$ and $\bar{\sigma}_j^{l_1} = h \cdot \sum_{i=1}^N u_{il_1} (x_{ij} - \bar{a}_j^{l_1})^2 / \sum_{i=1}^N u_{il_1}$ are two important parameters which have been generated by the FCM clustering algorithm [30], [43] in the previous sub-classifier, $l_1 = 1, 2, \dots, l'_{dp}$ and $j = 1, 2, \dots, d$.

Next, let us state how the remaining l''_{dp} interpretable fuzzy rules are generated. In order to assure the use of random rule heritage safely, we have two feasible strategies to generate the remaining l''_{dp} interpretable fuzzy rules. One is to first modify the classical clustering algorithm FCM [30], [43] with l_{dp} cluster centers into its variant where the l'_{dp} cluster centers are fixed in advance to achieve random rule heritage according to the IF-parts of inherited fuzzy rules. The resultant l''_{dp} cluster centers can then be assigned their linguistic values [16], [51] according to practical requirements. The other is to randomly generate the linguistically interpretable fuzzy partitions (e.g., 5 partitions respectively for *not high*, *little high*, *medium high*, *high* and *very high*) along each feature according to the same strategy as in [16], and then form the l''_{dp} interpretable IF-parts of fuzzy rules from these fuzzy partitions, which are the most apart from the l'_{dp} cluster centers fixed in advance according to the IF parts of inherited fuzzy rules. In this study, we prefer the latter for easy implementation.

According to the random generation strategy [16], we can get all the total l_{dp} interpretable IF-parts of fuzzy rules. In order to have random rule heritage with l'_{dp} fuzzy rules in the previous sub-classifier, we give up the total l'_{dp} IF-parts, which are as far as possible from the total l_{dp} IF-parts, and leave the total l''_{dp} ($=l_{dp} - l'_{dp}$) IF-parts for the generation of the remaining interpretable fuzzy rules in current sub-classifier.

After all the interpretable IF-parts of fuzzy rules are fixed in the above way in current sub-classifier, we compute the values of Gaussian membership function for the input sample $\mathbf{x}_i$ along the j -th feature [16].

$$u(l_2, x_{ij}) = \exp\left(-\left(x_{ij} - a_j^{l_2}\right)^2 / 2\left(\sigma_j^{l_2}\right)^2\right) \quad (8)$$

where $l_2 = 1, 2, \dots, l''_{dp}$, $i = 1, 2, \dots, N$, $j = 1, 2, \dots, d$.

We then compute the following values of the IF-parts of both the total l'_{dp} randomly inherited fuzzy rules and the total l''_{dp} newly generated fuzzy rules, respectively [16].

$$\bar{w}_{il_1} = \prod_{j=1}^d \bar{u}(l_1, x_{ij}); w_{il_2} = \prod_{j=1}^d u(l_2, x_{ij}) \quad (9)$$

where $\bar{w}_{il_1} \in \mathbf{R}^{d \times l'_{dp}}$, $w_{il_2} \in \mathbf{R}^{d \times l''_{dp}}$, $i = 1, 2, \dots, N$, $j = 1, 2, \dots, d$, $l_1 = 1, 2, \dots, l'_{dp}$ and $l_2 = 1, 2, \dots, l''_{dp}$. The rule layer output matrix $\mathbf{H}_{dp}$ can accordingly be constructed as follows: [(10) shown at the bottom of the next page]

As shown in Fig. 1, the solution of the output weight vector β_{dp} of the dp -th sub-classifier can be obtained by two steps. Firstly, we take the least learning machine (LLM), which had been described in detail in [18], [19], to get the analytical solution of the output weight vector β_1 of the first sub-classifier. Secondly, we describe how to find the output weight vector β_{dp} (i.e., consequent parameter vector β_{dp} in fuzzy rules) in each remaining training sub-classifier (i.e., $dp \geq 2$). In this study, in order to solve β_{dp} analytically, we simply change LLM's objective function (LLM's variant in [18], [19]) into the following squared error function according to the similar strategy as in [50]:

$$\Psi(\beta_{dp}) = \|(1 + \lambda')\mathbf{H}_{dp}\beta_{dp} - (\mathbf{T} - (1 - \delta')\mathbf{Y}_{dp-1})\|^2 \quad (11)$$

where $\mathbf{H}_{dp}$ and β_{dp} are $N \times l_{dp}$ and $l_{dp} \times c$ matrices, respectively. $l_{dp} = l'_{dp} + l''_{dp}$. $\lambda' < 0$ and $\delta' > 0$ are the given regularization parameters which are used to characterize different roles of two adjacent training sub-classifiers. The rationality of (11) can be stated as follows. Intuitively, β_{dp} should be more dependent on current training sub-classifier than on the previous training sub-classifier. That is to say, when $\lambda' \neq 0$ and/or $\delta' \neq 0$, according to (11), the outputs of current training sub-classifier may be over-emphasized, while the outputs of the previous training sub-classifier may be under-emphasized. In particular, when $\lambda'=0$ and $\delta'=0$, (11) degenerates into the usual squared error function [18], [19]:

$$\Psi(\beta_{dp}) = \|\mathbf{H}_{dp}\beta_{dp} - (\mathbf{T} - \mathbf{Y}_{dp-1})\|^2 \quad (12)$$

At last, referring to [16], the output matrix $\mathbf{Y}_{dp}$ can be calculated as follows.

$$\mathbf{Y}_{dp} = \mathbf{H}_{dp}\beta_{dp} \quad (13)$$

Part II—On linear combination of all sub-classifiers: According to the stacked fuzzy classifiers, it seems that we may simply take the output of the last sub-classifier as the final output of the entire deep training model. However, let us recall that an obvious characteristic of sleep-wake stage data is its changing importance of the features caused by the uncertainties over time. While the input space of each sub-classifier is generated by the original input space plus the randomly projected outputs of the previous sub-classifier, it implicitly means the injection of uncertainties in the input space of each sub-classifier. As a result, we should take a combination of all sub-classifiers to guarantee better overall classification accuracy according to the same strategy as in [50]. After considering the characteristic of stacked learning in the proposed structure, with our extensive experimental justification, we take the following easy yet more efficient linear combination strategy, instead of the “majority voting” or “average” ensemble strategy. Assume the outputs of all sub-classifiers are expressed respectively as

$f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_{DP}(\mathbf{x})$ for the input sample $\mathbf{x}$, the final output $f(\mathbf{x})$ of the proposed classifier can be expressed as

$$f(\mathbf{x}) = \sum_{j=1}^{DP} \lambda_j f_j(\mathbf{x}) \quad (14)$$

where $\lambda_1, \lambda_2, \dots, \lambda_{DP}$ are these combination coefficients. Below we will state how to determine these appropriate parameters $\lambda_1, \lambda_2, \dots, \lambda_{DP}$ in detail.

Let y denote the corresponding actual class label of input sample $\mathbf{x}$. For the given training set, we consider three factors: (1) $f(\mathbf{x})$ should approximate the corresponding actual class label as well as possible; (2) the output of the first sub-classifier and the corresponding actual class label can be regularized to have their mutual approximation; (3) from the second sub-classifier, the output of each sub-classifier should approximate the maximal output of the previous sub-classifiers as well as possible. As a result, the following training error is defined by regularizing these three factors according to the similar strategy as in [50]:

$$\begin{aligned} J(\lambda_1, \lambda_2, \dots, \lambda_{DP}) &= \sum_{i=1}^N \left[(\lambda_1 f_1(\mathbf{x}_i) - \xi y_i) + \sum_{j=2}^{DP} (\lambda_j f_j(\mathbf{x}_i) - \max\{f_1(\mathbf{x}_i), f_2(\mathbf{x}_i), \dots, f_{j-1}(\mathbf{x}_i)\}) \right. \\ &\quad \left. + \zeta \left(\sum_{j=1}^{DP} \lambda_j f_j(\mathbf{x}_i) - y_i \right) \right]^2 \\ &= \sum_{i=1}^N \left[\left(\sum_{j=1}^{DP} \lambda_j f_j(\mathbf{x}_i) - \left(\xi y_i + \sum_{j=2}^{DP} \max\{f_1(\mathbf{x}_i), f_2(\mathbf{x}_i), \dots, f_{j-1}(\mathbf{x}_i)\} \right) \right) \right. \\ &\quad \left. + \zeta \left(\sum_{j=1}^{DP} \lambda_j f_j(\mathbf{x}_i) - y_i \right) \right]^2 \\ &= \sum_{i=1}^N \left[(1 + \zeta) \sum_{j=1}^{DP} \lambda_j f_j(\mathbf{x}_i) - (\xi y_i + \zeta y_i + \sum_{j=2}^{DP} \max\{f_1(\mathbf{x}_i), f_2(\mathbf{x}_i), \dots, f_{j-1}(\mathbf{x}_i)\}) \right]^2 \end{aligned} \quad (15)$$

where ξ and ζ are two regularization coefficients which can be pre-set and manually adjusted. (15) can be equivalently

expressed as

$$J(\lambda_1, \lambda_2, \dots, \lambda_{DP}) = \sum_{i=1}^N [(1 + \zeta) \boldsymbol{\lambda}^T \mathbf{F}_i - \boldsymbol{\sigma}_i]^2 \quad (16)$$

where $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_{DP})^T$, $\mathbf{F}_i = (f_1(\mathbf{x}_i), f_2(\mathbf{x}_i), \dots, f_{DP}(\mathbf{x}_i))^T$ and $\boldsymbol{\sigma}_i = \xi y_i + \zeta y_i + \sum_{j=2}^{DP} \max\{f_1(\mathbf{x}_i), f_2(\mathbf{x}_i), \dots, f_{j-1}(\mathbf{x}_i)\}$. By setting the derivatives with respect to $\boldsymbol{\lambda}$ to be zero [50], we have

$$\partial J(\lambda_1, \lambda_2, \dots, \lambda_{DP}) / \partial \boldsymbol{\lambda} = 0 \quad (17)$$

We can readily have their analytical solution, i.e.,

$$\boldsymbol{\lambda}^T = \frac{1}{1 + \zeta} \left(\sum_{i=1}^N (\boldsymbol{\sigma}_i \mathbf{F}_i^T) \right) \left(\sum_{i=1}^N (\mathbf{F}_i \mathbf{F}_i^T) \right)^{-1} \quad (18)$$

Part III—On deep learning algorithm: Next we give the proposed deep learning algorithm with random rule heritage in part 1 of Section III-B-1. We also have the following remarks about the above algorithm.

Remark 1: The total five parameters involved in the above learning algorithm should be appropriately set. Concretely, according to our extensive experiments, two regularization coefficients ξ and ζ can be pre-set in [01]. Owing to the use of deep structure in Drrh-TSK-FC, our extensive experimental results indicate that for various datasets, the total number of rules in each sub-classifier is generally comparatively small, i.e., ranging from 5 to 15. 10% to 30% and 5% to 10% of the total number of rules in the $(dp-1)$ -th sub-classifier are taken respectively for the use of fuzzy random rule heritage (as l'_{dp}) and short fuzzy rules. The total number DP of sub-classifiers generally ranges from 2 to 6.

Remark 2: Let us observe the time complexity of the above training process of the proposed classifier Drrh-TSK-FC. Referring to the time complexity analysis of our recent work in [16], the first sub-classifier has the same training as that in [16], thus its training time complexity is $O(5Nd^2l_1 + l_1^3)$. Due to the use of random rule heritage, the training of all the remaining sub-classifiers is slightly different from those in [16]. However, in terms of steps 4-5 in Part II, we can derive that the time complexity of training each sub-classifier ($dp > 1$) is roughly $O(5Nd^2(l' + l'') + (l' + l'')^3)$. As a result, the entire time complexity of Drrh-TSK-FC roughly becomes $O(5Nd^2(l_1 + \sum_{dp=2}^{DP} (l'_{dp} + l''_{dp})) + l_1^3 + \sum_{dp=2}^{DP} ((l'_{dp} + l''_{dp})^3))$. Obviously, when both the total number

$$\mathbf{H}_{dp} = \begin{bmatrix} \underbrace{\bar{w}_{11} \quad \cdots \quad \bar{w}_{1l'_{dp}}}_{\text{random-rule-heritage}} & \underbrace{w_{1(l'_{dp}+1)} \quad \cdots \quad w_{1(l'_{dp}+l''_{dp})}}_{\text{generated-rules}} \\ \vdots & \vdots \\ \underbrace{\bar{w}_{N1} \quad \cdots \quad \bar{w}_{Nl'_{dp}}}_{\text{random-rule-heritage}} & \underbrace{w_{N(l'_{dp}+1)} \quad \cdots \quad w_{N(l'_{dp}+l''_{dp})}}_{\text{generated-rules}} \end{bmatrix}_{N \times (l'_{dp} + l''_{dp})} \quad (10)$$

Algorithm: Deep Learning Algorithm of Drrh-TSK-FC.

Initialization:

According to the training of the first sub-classifier,
initialize

$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^T$, $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_N]^T$,
regularization coefficients ξ

and ζ , the total number l_{dp} of rules in each sub-classifier,
 l'_{dp}

for fuzzy rule heritage proportion and DP in Part I and
then
obtain $\mathbf{Y}_1$;

Training the remaining sub-classifiers:

while $dp \leq DP$ do

Step 1: Fix l_{dp} and randomly generate the l_{dp} cluster
centers and finish the random rule heritage with
the total l'_{dp} IF-parts

Step 2: Compute $\bar{u}(l_1, x_{ij})$ and $u(l_2, x_{ij})$

Step 3: Compute $\bar{w}_{il_1}$ and w_{il_2}

Step 4: Construct the rule layer output matrix $\mathbf{H}_{dp}$

Step 5: Compute the output weight vector β_{dp} and the
output

matrix $\mathbf{Y}_{dp}$ according to LLM in [18]

Step 6: Update $\mathbf{X}_{dp+1}$ according to [16]

end while

Output the total DP sub-classifiers, and the combination
coefficients $\lambda_1, \lambda_2, \dots, \lambda_{DP}$.

of fuzzy rules and the total number d of features are comparatively small, it is still linear with a considerably big N .

IV. EXPERIMENTAL STUDIES

In order to observe the performance of Drrh-TSK-FC, it is evaluated with other seven classifiers (i.e., zero-order & first-order TSK fuzzy classifier [16], [17], Libsvm (Linear & Gaussian) [31], C4.5 [32], D-TSK-FC [16] and DBN [33]) on sleep stage data from MIT-BIH Polysomnographic (PSG) database [34]. The average training/testing classification accuracies in mean (i.e., which are denoted as $Trac$ and $Teac$ in the corresponding experimental results, respectively) are taken for performance evaluation for all comparative classifiers. In addition, the total accuracy (i.e., $ACC = \frac{TP+TN}{TP+FN+FP+TN}$), sensitivity (i.e., $SEN = \frac{TP}{TP+FN}$) and specificity (i.e., $SPE = \frac{TN}{TN+FP}$) are also used to evaluate the performance of all the classifiers according to [35], [42] where TP , TN , FP and FN denote true positives, true negatives, false positives and false negative, respectively. All the experiments are based on the MIT-BIH polysomnographic database in PhysioBank archives (<https://physionet.org/physiobank/database/slpdb/>). All the adopted classifiers are carried out with MATLAB R2016a on a computing platform with Core (TM) i3-4150 3.5 GHZ CPU with 4 GB memory. In our experiments, each dataset is normalized and then randomly partitioned such that 75% of the data are taken for training and the remaining for testing. The experimental studies are detailed as follows.

A. On Data Preparation

Here we describe how to generate sleep-wake stage data from the original EEG signals downloaded from <https://physionet.org/physiobank/database/slpdb/>.

1) *Preprocessing*: Due to the existence of noises and uncertainties in sleep-wake stage data, an ideal classification methodology on such data should be robust in the sense of noise removal, uncertainty handling and generalization performance. In order to effectively tackle with noises and uncertainties in sleep-wake stage data, except for robust classification methods, a necessary preprocessing step should generally be taken to have considerable removal of noises in the original signals. Male signals (including ECG signals, blood pressure signals, electroencephalogram signals, respiratory signals, etc.) aged from 32 to 56 and weighing from 89 to 152 kg are used in this study [34]. The frequency of these signals basically concentrates between 0.5 and 30 Hz. In our experimental studies, we use a bandpass filter to retain the effective frequency signal between 0.5 and 30 Hz and eliminate the high-frequency noise signal with many interference [34]. Due to the existence of various high frequency noises in the original EEG signal, the signal waveform is disordered [34]. We use a Butterworth bandpass filter to filter the EEG signal. For example, taking slp02a as an example, we intercept and filter the first 1000 sample points to ensure that the signal frequency is concentrated within 0.5~30 Hz and the signal waveform is smooth. However, there is still noise in the EEG signal. In this experiment, DB6 wavelet basis function [34] is used to further denoise the EEG signal, and global threshold is used to quantify the high frequency coefficient, and the signal and noise are separated. Please note, there may still be some degree of uncertainties after preprocessing sleep-wake stage data. As pointed out in [14], [15] and the introduction section, fuzzy systems have strong uncertainty-handling capability, and deep fuzzy structures have strong generalization capability. Therefore, the proposed Drrh-TSK-FC naturally has a great potential for robust classification of sleep-wake stage data.

2) Feature Extraction and Generation of Sleep Stage Datasets

We use Gaussian kernel principal component analysis (KPCA) to extract efficient features from 7500 features acquired after preprocessing [34]. According to the EEG signals obtained by KPCA, following [1], [34], we randomly combine 18 polysomnography segments (i.e., slp01a, slp01b, slp02a, slp02b, slp03, slp04, slp14, slp16, slp32, slp37, slp41, slp45, slp48, slp59, slp60, slp61, slp66 and slp67x) corresponding to each signal to randomly generate a dataset. In this way, we generate four datasets of sleep-wake stages, i.e., *Data-1*, *Data-2*, *Data-3* and *Data-4* in Table I, for the diversity of experimental studies. In addition, we follow the same arrangement as in [46], [47] to separate a dataset into its training and testing data for fair comparison.

B. On Comparative Classifiers and Parameter Settings

In our experiments, Drrh-TSK-FC is compared with 5 swallow classifiers (i.e., zero-order & first-order TSK fuzzy classifiers [16], [17], Libsvm (Linear & Gaussian) [31], C4.5 [32]) and

TABLE II
PARAMETER SETTINGS FOR ALL THE COMPARATIVE METHODS

#classifiers	parameter settings
Zero-order & first-order classifiers	regularization parameter and scale parameter ranging from 0.01 to 100 with the interval being 0.1; the number of clusters being equal to the number of fuzzy rules;
Libsvm	kernel width chosen from $\{0.1, 1, 5, 10, 20, 50, 100, 150, 200\}$;
D-TSK-FC	the numbers of fuzzy rules ranging from 4 to 20 with the interval being 1; $DP=3$;
DBN with 3-layer restricted Boltzmann machines (RBM) (*)	the numbers of hidden nodes ranging from 4 to 20 with the interval being 1 for <i>Data-1</i> , <i>Data-2</i> and <i>Data-3</i> ; from 4 to 15 with the interval being 1 for <i>Data-4</i> ; $maxepoch=50$, $weightcost=0.0002$, $epsilon_{lw}=0.1$, $epsilon_{lvb}=0.1$, $epsilon_{hnb}=0.1$, $initialmomentum=0.5$ and $finalmomentum=0.9$;
C4.5	the numbers of inc_nodes and regions being 5 and 10, respectively;
Drrh-TSK-FC with $DP=3$ or $DP=4$	the number of fuzzy rules in each sub-classifier ranging from 5 to 15 with the interval being 1; $\xi, \zeta, \epsilon \in [0, 1]$; i_{dp} for fuzzy rule heritage ranging from 10% to 30% of the total number of rules in the dp th sub-classifier ($dp > 1$); 5% to 10% of the total number of rules for the use of short fuzzy rules; the training depth DP ranging from 2 to 6;

(*) $maxepoch$ represents the maximum number of iterations; $weightcost$ is the coefficient of error; $epsilon_{lw}$, $epsilon_{lvb}$ and $epsilon_{hnb}$ represent the learning rate of the weight and the input layer bias and the hidden layer, respectively; $initialmomentum$ and $finalmomentum$ represent the amplitude coefficients of the corresponding update bias at less than or greater than 5 iterations.

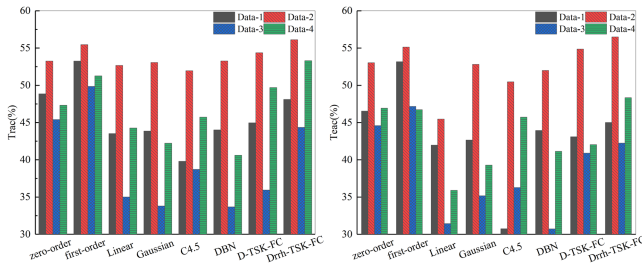


Fig. 2. Tracs/teacs (%) about all classifiers.

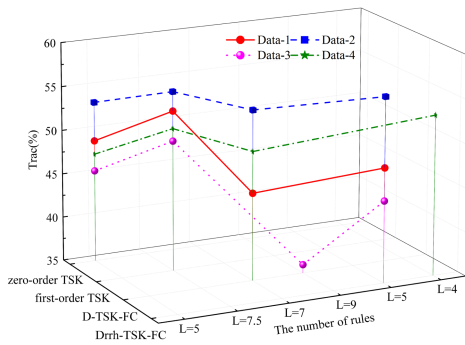


Fig. 3. The number of fuzzy rules about all fuzzy classifiers.

2 deep classifiers (*i.e.*, D-TSK-FC [16] and DBN [33]). Table II describes the parameter settings in detail.

C. On Results and Discussions

Figs. 2 and 3 show respectively *Tracs* and *Teacs* of all the comparative classifiers and the numbers of fuzzy rules by the corresponding fuzzy classifiers on the adopted four *datasets*. For *Data-1*, *Data-3* and *Data-4*, *Tracs* of Drrh-TSK-FC are 48.11%,

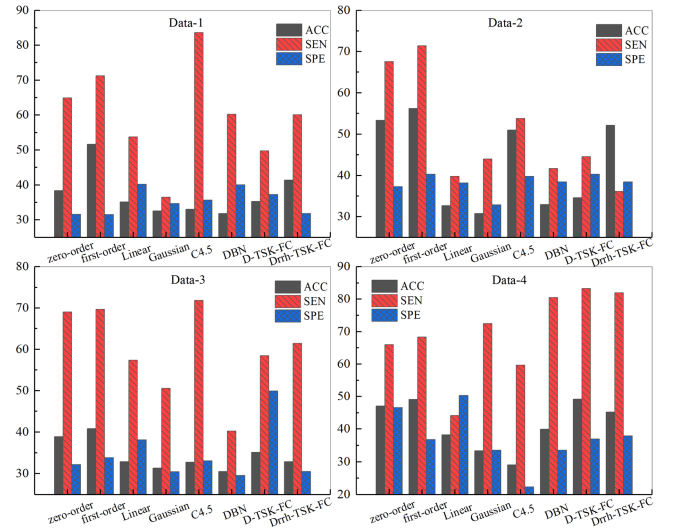


Fig. 4. ACC, SEN and SPE (%) about all classifiers.

44.36% and 53.31%, respectively. Obviously, Drrh-TSK-FC wins the other five comparative classifiers, but is slightly worse than the zero-order & first-order TSK fuzzy classifier in most datasets. In addition, compared with all the other comparative classifiers, Drrh-TSK-FC achieves at least comparable *Tracs* for *Data-2*. We can also find that among these *Teacs*, Drrh-TSK-FC maintains satisfactory results, which actually means that Drrh-TSK-FC has better generalization capability. Compared with all the other fuzzy classifiers, Drrh-TSK-FC takes smaller or comparable number of fuzzy rules according to Fig. 3. Therefore, we can think of Drrh-TSK-FC as having good interpretability.

One-against-all-rest strategy is taken to objectively evaluate the performance of each classifier in this experiment. Besides, we report the mean of each evaluation indices on all sleep stages. Fig. 4 lists the results of evaluation indices (*i.e.*, *ACC*, *SEN* and *SPE*) of all the comparative classifiers on all datasets. As for the average values of *SEN*, Drrh-TSK-FC are also satisfactory, which means that Drrh-TSK-FC is better than most of all the comparative classifiers in identifying positive classes. However, we also find that the average values of *SEN* by Drrh-TSK-FC is worse than that of the other classifiers on *Data-1*, *Data-2* and *Data-3*. In addition, according to the average values of *ACC* and *SPE* obtained by Drrh-TSK-FC, they seem to be not so high. However, we can easily find that the classification results of Drrh-TSK-FC do not violate the existing experimental consensus in [45] and keep at least comparable to most of all the other comparative classifiers on these datasets. In fact, when Drrh-TSK-FC has its interpretability guarantee, the fact that it still maintains such classification results reasonably indicates the robustness of Drrh-TSK-FC in the sense of both uncertainty-handling and generalization capability coming from fuzzy sub-classifiers and their deep structure.

Therefore, we can believe that Drrh-TSK-FC basically performs well. However, likewise in other comparative classifiers, Drrh-TSK-FC does not work very well on some datasets which are difficult to extract features. Even so, we should keep in mind

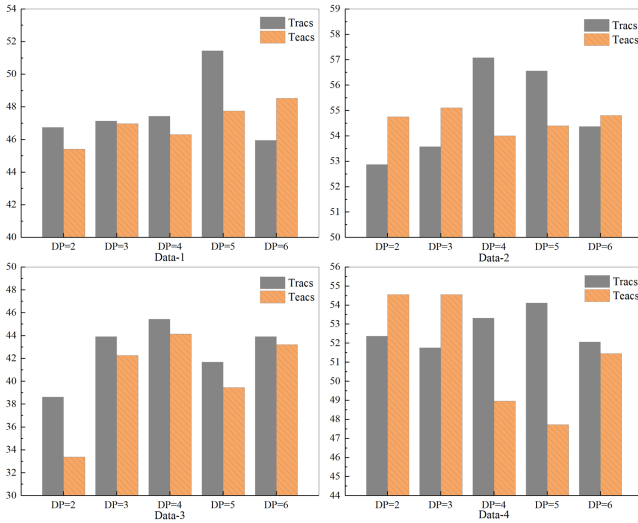


Fig. 5. Tracs/teacs (%) of Drrh-TSK-FC with different heritage features.

that Drrh-TSK-FC has its advantage in the sense of interpretability. In addition, we may design better feature extraction methods to deal with these complex classification data in the future.

D. On Impact of Random Rule Heritage on Drrh-TSK-FC

In this subsection, we report how the classification performance of Drrh-TSK-FC is affected by random rule heritage from two observational aspects. One is the number of random heritage rules from the last sub-classifier to current sub-classifier. The other observational dimension is the proportion of rules randomly inherited from the last sub-classifier to current sub-classifier.

E. Impact of the Number of Random Heritage Rules

In this section, we discuss the relationship between the classification performance of Drrh-TSK-FC and random rule heritage among sub-classifiers. We can clearly see the *Tracs/Teacs* of Drrh-TSK-FC after using random rule heritage according to Fig. 5.

Generally speaking, the classification performance of Drrh-TSK-FC is basically optimal after one or four rule heritages. The training depth (the number of sub-classifiers) of Drrh-TSK-FC is set to be from 2 to 6 in this experiment to obtain satisfactory classification performance. Next, we will make a detailed analysis of Figs. 2 and 5.

According to *Data-1* in Fig. 5, we find that the classification performance of Drrh-TSK-FC is superior after four heritage-rule sub-classifiers, that is, the *Tracs* (*Teacs*) increase from 46.74% (45.41%) to 51.43% (47.75%). In addition, the initial *Tracs* and *Teacs* of Drrh-TSK-FC are higher than that of most other classifiers. In other words, this rule heritage method actually widens the gap between Drrh-TSK-FC and other classifiers. For *Data-2*, the initial *Trac* of Drrh-TSK-FC is 52.87%, while zero-order TSK fuzzy classifier, first-order TSK fuzzy classifier, C4.5, DBN and D-TSK-FC are 53.26%, 55.46%, 51.97%,

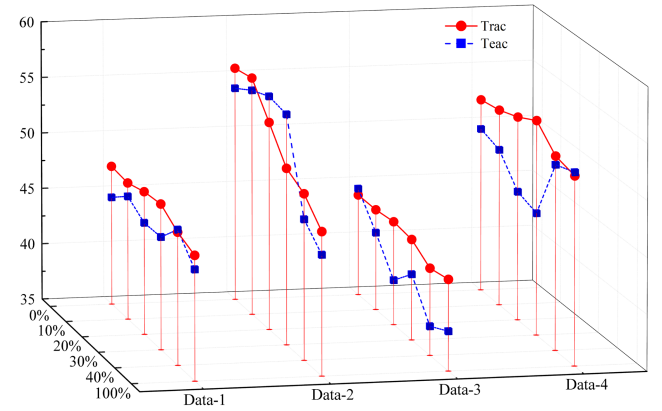


Fig. 6. Training/testing accuracies with the different proportions of random heritage rules.

53.27% and 54.37%, respectively. Therefore, the classification performance of Drrh-TSK-FC with two heritage rules maintains comparable classification results with first-order TSK fuzzy classifier mentioned above. After introducing the rule heritage, we can see that the *Trac* is improved to 57.08% after the two heritage-rule sub-classifiers which enhances the *Trac*. For *Data-3* and *Data-4*, the classification performance of Drrh-TSK-FC has been basically improved after two/three heritage-rule sub-classifiers. Compared with other most classifiers mentioned above, Drrh-TSK-FC has great advantages in classification performance. Therefore, this unique random heritage rule really helps Drrh-TSK-FC achieve the goal of improving classification performance.

F. Impact of the Proportion of Randomly Inherited Rules

This subsection further analyses the impact of randomly inherited rules on the classification performance of Drrh-TSK-FC. In order to show more intuitively the relationship between the number of randomly inherited rules and the classification performance of Drrh-TSK-FC, 10%, 20%, 30% and 40% of the rules from current sub-classifier are randomly selected to inherit to the next sub-classifier, and then observe the change of their classification performance. Fig. 6 shows the classification performance of Drrh-TSK-FC for different proportions of randomly heritage rules. According to Fig. 6, it is not difficult to summarize the followings.

- Generally speaking, the classification performance of Drrh-TSK-FC shows a slightly increased or slightly decreased trend with the proportion of random rule heritage. This phenomenon is particularly evident, as listed in Fig. 6. Fig. 6 further validates our previous prediction that the classification performance of Drrh-TSK-FC increases or slightly decreases with the increase of random heritage rules. The reason is that random heritage rules already contain some features or classification information of input samples. These features or classification information play a role in promoting and guiding the classification of next base-sub-classifier. With the increasing number of

TABLE III
LINGUISTIC EXPLANATIONS IN IF-PART AND THEN-PART OF ONE FUZZY RULE OBTAINED ON DATA-1

Rule #:								
IF-part	the energy of the EEG signal in							
	Band 1		Band 2		Band 3		...	
	$a_1^2 = 0.5 \in \Omega_3$	$\sigma_1^2 = 0.1871$	$a_2^2 = 1.0 \in \Omega_5$	$\sigma_2^2 = -1.6855$	$a_3^2 = 0.75 \in \Omega_4$	$\sigma_3^2 = -0.6319$	...	...
	the corresponding linguistic interpretability							
	<i>medium serious</i>		<i>especially serious</i>		<i>more serious</i>		...	
	$p_{20} = +0.9331$							
THEN-part								

random heritage rules, the situation of improving the classification performance of Drrh-TSK-FC gradually slows down or even degrades the classification performance. Therefore, with the increase of the number of random heritage rules, the computational complexity of the training algorithm will increase accordingly, which will inevitably lead to the weakening of the interpretability of Drrh-TSK-FC.

- b) Now we discuss the effects of two extreme cases on the classification performance of Drrh-TSK-FC. The first case is that no any rule is randomly inherited (i.e., the proportion of randomly inherited rules is 0). The second case is to inherit all rules (i.e., the proportion of heritage rules is 100%). According to Fig. 6, we find two interesting phenomena. (1) Compared to Drrh-TSK-FC with the first case, the *Tracs/Teacs* of Drrh-TSK-FC with the use of random rule heritage shows a slight increase, which indicates that the random rule heritage can be helpful in improving the classification performance of Drrh-TSK-FC. (2) The *Tracs/Teacs* of Drrh-TSK-FC are basically stable or even slightly declining when all rules are inherited. This also shows that the random rule heritage can enhance the classification performance.

G. On Parameter Sensitivity

According to the above analysis, we know that the performance of Drrh-TSK-FC heavily depends on three factors: the training depth DP , the proportion ω of random heritage rules and the number L of fuzzy rules. In order to more intuitively observe the impact of these three factors on *Tracs*, we keep two of them unchanged and then observe the effect of the third factor on the best *Trac* of Drrh-TSK-FC in a one-by-one way.

Fig. 7 illustrates the impact of three factors on the performance of Drrh-TSK-FC on all *Datasets*. According to Fig. 7, we can easily see: (1) the *Tracs* obviously increase when the number of fuzzy rules increases from 4 to 8 or from 4 to 11. If the fuzzy rules are consecutively increased, the *Tracs* will basically remain stable or show a slight downward trend. (2) the *Tracs* keep increasing with the increase of the number of random heritage rules from 10% to 30% in *Data_2* and *Data_4*, and keep slightly decreasing with the increase of the number of random heritage rules from 30% to 40%, which is consistent with the conclusion we discussed above. (3) From the experiments we arranged here, the satisfied *Tracs* can be achieved when DP reaches 3 or 4 and

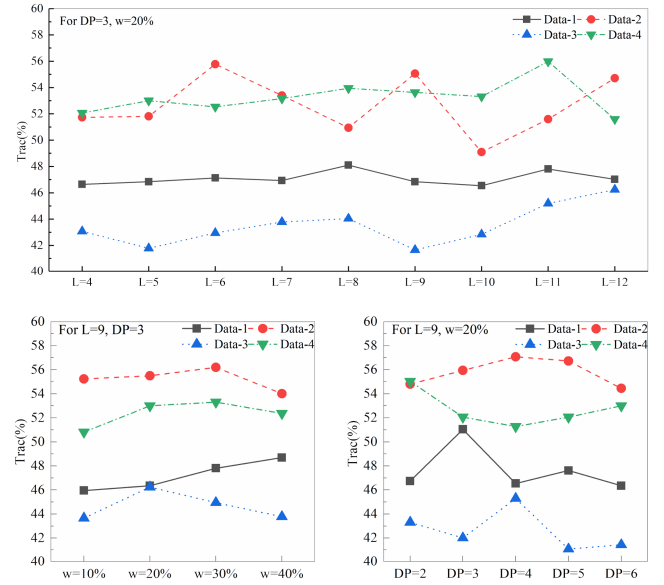


Fig. 7. Impact of three factors on classification performance of DRRH-TSK-FC on all datasets.

the number of fuzzy rules and the proportion of random heritage rules are set to 30% or 40%. Bigger depths and more heritage rules may perhaps yield overfitting phenomenon. Therefore, how to choose a suitable training depth and/or heritage rules has a decisive impact on classification performance.

H. On Interpretability of Drrh-TSK-FC

In this section, we observe the interpretability of Drrh-TSK-FC. Limited by the length of the paper, we only list randomly one of all the fuzzy rules by Drrh-TSK-FC on *Data-1* (as shown in Table III). We will see that this rule can be linguistically understandable. Concretely, the antecedent parts (denoted by the clustering centers and variances) of the first fuzzy rule are (0.5, 0.1871), (1.0, -1.6855), (0.75, -0.6319) and so on. In order to assign linguistic values conveniently, we divide $[0, 1]$ into five equal intervals, namely $\Omega_1 = [0, 0.2)$, $\Omega_2 = [0.2, 0.4)$, $\Omega_3 = [0.4, 0.6)$, $\Omega_4 = [0.6, 0.8)$ and $\Omega_5 = [0.8, 1.0]$. Obviously, when the clustering centers fall in these five intervals, the corresponding fuzzy membership functions may be assigned with the corresponding linguistic values: *not serious*, *little serious*, *medium serious*, *more serious* and *especially serious*, respectively.

V. CONCLUSION

Due to the existence of noise and uncertainty in sleep-wake stage, this study develops a deep TSK fuzzy classifier Drrh-TSK-FC to achieve an efficient sleep stage recognition with high interpretability. On one hand, Drrh-TSK-FC keeps the same generation of input spaces in its sub-classifiers as in our previous works such that it can well share the distinctive advantage of deep stacked structures in the sense of enhanced performance and interpretability. On the other hand, different from traditional deep stacked structures, Drrh-TSK-FC considers the use of both short fuzzy rules and random rule heritage in its each sub-classifier so as to reflect the importance changes of the features in adjacent sub-classifiers and hence mimic human cognitive behavior in leveraging the past experiences to address similar problems. Besides, Drrh-TSK-FC takes a linear ensemble of all the sub-classifiers to assure further improvement in its performance. The experimental results show that Drrh-TSK-FC is superior to other comparative classifiers for the tasks of sleep-wake stage recognition.

There is a big room for the future study about this work here. For example, how to determine an appropriate number of randomly inherited rules or give a guideline about it for various datasets is worthy for study in near future. While one may argue that the importance changes of features should be better expressed by their different weights rather than the use of short fuzzy rules, how to generalize the use of short fuzzy rules is another interesting topic. At present, our study only concentrates on a linear combination of all the sub-classifiers in Drrh-TSK-FC. Let us recall that another two commonly used ensemble strategies, i.e., averaging and majority voting, generally require much less computational burden. Therefore, which ensemble strategy should be taken by Drrh-TSK-FC so as to achieve good trade-off between classification performance and computational burden should be empirically studied for various datasets. In near future, we will go on our studies from these perspectives.

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